

*"Navigating the Pulse of Jakarta": **Uncovering Hidden Patterns Behind Millions of Transjakarta Transactions***

Irfan Maulana

Data Analyst – Final Project Ms. Excel DØLab



Business Overview Transjakarta

- 1. The Pulse of Mobility:** Transjakarta is not just a mode of transportation; it is the pulse of Jakarta citizens' mobility with millions of daily transactions.
- 2. World's Longest BRT:** Transjakarta is not merely a city bus, but a Bus Rapid Transit (BRT) system with the longest track in the world (over 250 km).
- 3. Integration Ecosystem (JakLingko):** Currently, Transjakarta has transformed into a giant ecosystem under the umbrella of JakLingko, integrating Big Buses (BRT), Non-BRT Buses, and micro-transport (Mikrotrans/Angkot) with a single centralized cashless payment system.
- 4. Massive Volume:** Serving millions of customers daily, generating transaction data that is extremely large (Big Data) and complex.

Storytelling Dataset Transjakarta

Every time a card is tapped at the gate (Tap-In), it is not just a ticket payment. It is a digital footprint telling the story of Jakarta citizens' lives.

This dataset records passenger activity for 1 full month, covering:

- **Who are they?** (Demographic profile & payment preferences).
- **When do they move?** (Peak hour patterns and holidays).
- **Where do they go?** (Favorite routes and crowd centers).
- **What is the economic value?** (Revenue contribution per hour)

Strategic Missions

The objectives of this analysis are summarized in 3 Strategic Pillars:

- **Revenue Optimization:** Identifying key revenue sources and validating the Pareto Law (80/20) in operations.
- **Operational Excellence:** Adjusting fleet schedules based on accurate Peak Hour data (not just intuition).
- **Customer Centricity:** Understanding passenger profiles (Generation & Bank) for more personalized partnership and promotion strategies.



transjakarta

Data Proses di Power Query

Process data to get insight

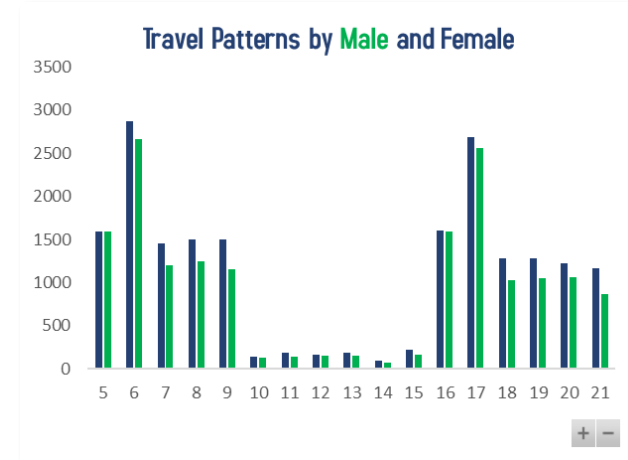
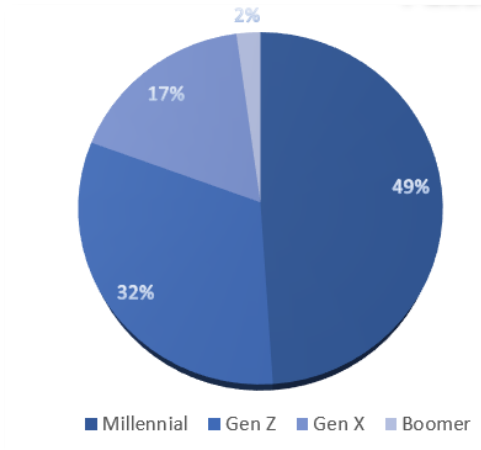
Transjakarta Overview (April 2023)



- **Total Revenue:** Rp94,628,000.
- **Total Transactions:** 35,970 passengers.
- **Average Ticket Price:** Rp2,702.73.
- **Missing Tap-out:** 1,268 (approx. 3.53% of passengers failed to tap out).

Passenger Demographics

- Passengers are dominated by **Millennials** (49%), followed by Gen Z (32%). These groups are the working class requiring high mobility at affordable prices.
- Interestingly, Transjakarta passengers are predominantly Female compared to Male.



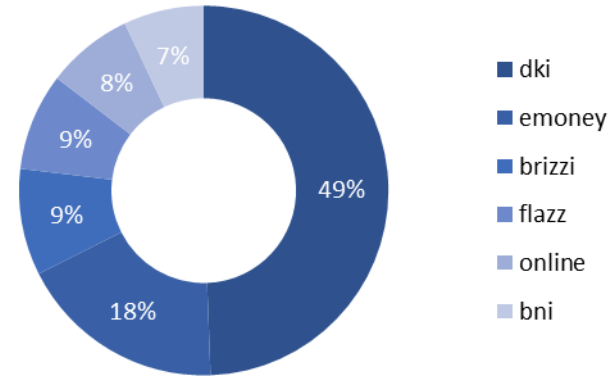
Insight:

- Since 80% of passengers are **Digital Natives**, communication strategies (complaints, route info) must be **digital-first** (App/Social Media). Do not spend budget on printed brochures.
- Since the majority are women, "**Anti-Sexual Harassment**" programs and CCTV/Officers on buses are key to customer retention. If women feel unsafe, Transjakarta risks losing its largest customer base.

Revenue by Payment Method

- Almost 50% of passengers use the **Bank DKI (JakCard)** ecosystem. This correlates strongly with Subsidy Recipients & Government Programs (Elderly, Students, Civil Servants).
- If combined, commercial banks (**Mandiri, BRI, BCA, BNI**) control about **43%** of the share
- Interestingly, the adoption of digital systems is growing but not yet dominant. This indicates that passengers are still more comfortable with 'Physical Card Tapping,' which is perceived as faster than having to access their phones first.

Revenue by Payment Method

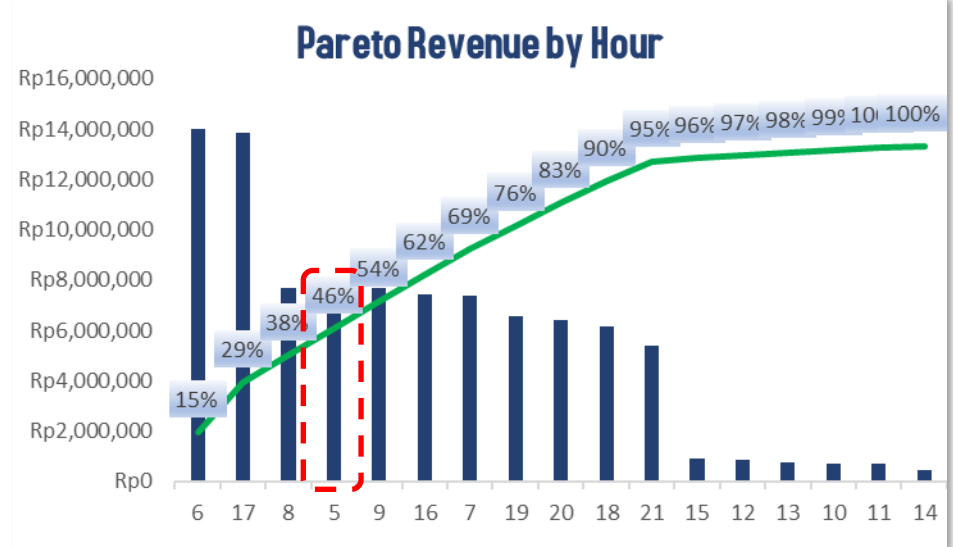


Insight:

- To increase Revenue (not just passenger count), marketing focus must be directed at the **43% commercial bank users**.
- There is huge untapped potential in **Bank BCA**, the largest retail transactional bank in Jakarta, to convert its users to bus riders.

The Golden 5 Hours

- In just **5 Operational Hours** (05:00–09:00 & 17:00), Transjakarta secures more than **50% of Daily Revenue**.
- 06:00 AM and 05:00 PM are the peak times, contributing nearly 30% of total revenue. Surprisingly, **05:00 AM** generates more revenue than 07:00 AM.

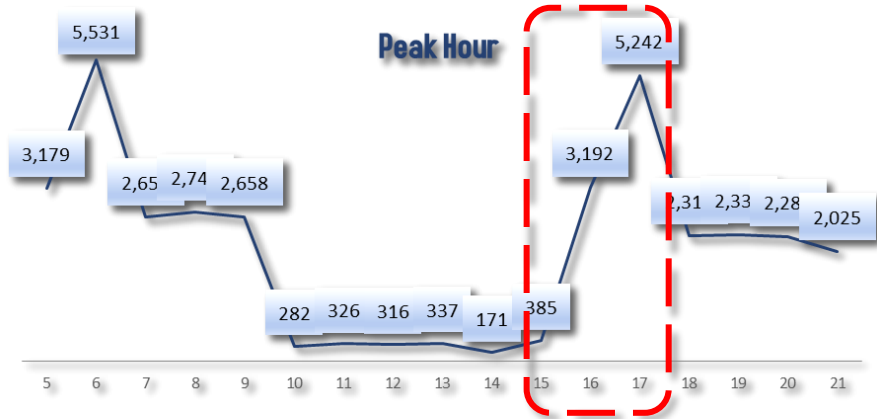


Insight:

- Deploy premium fleets at **05:00–06:00 AM** due to its high revenue contribution.
- Reduce frequency (loosen headway) during **10:00 – 14:00** to save fuel and costs, as the impact on lost revenue is minimal.

Peak Hour

- Transjakarta passengers are primarily "Long-Distance Commuters." They must depart very early (5–6 AM) from satellite cities (Bekasi, Depok, Tangerang) to reach Jakarta's central offices by 8 AM. Passengers traveling between 7–8 AM represent the "second wave" who live closer to the city center.
- A rapid surge occurs starting from 15:00 (385 passengers), exploding to a peak at 17:00 (5,242 passengers).
- An extreme drop in traffic occurs during mid-day hours (10:00 AM – 2:00 PM), with an average of only 200–300 passengers.

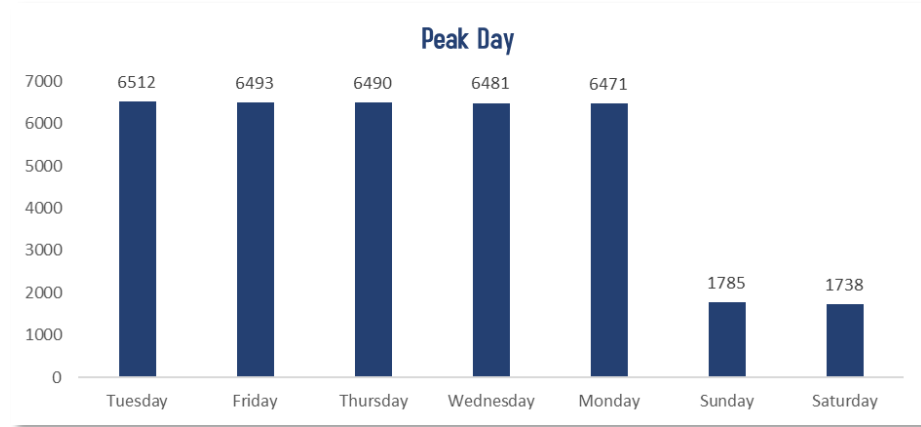


Insight:

- Apply **Shift Schedules**. Operate 100% fleet at 05:00–09:00, then pull 70% of the fleet to the pool at 10:00–14:00 for maintenance/rest.

Peak Day

- There is almost 0% variance from Monday to Friday. Passengers are the same people with the same routine every workday.
- When offices close, Transjakarta loses **3/4 of its customers**. This proves it is not yet a top choice for leisure/tourism.
- Sunday traffic is slightly higher than Saturday, likely due to Car Free Day (CFD) or social events.



Insight:

- Drastically cut fleet operations on weekends (73% traffic drop) to prevent losses.
- Create a **"Cheap Jakarta Tour"** campaign specifically for weekends to boost ridership.

Busiest Corridors

- **Cibubur - Balai Kota** is the busiest because KRL (train) access is not as convenient there compared to Bekasi/Serpong. They are highly dependent on Transjakarta.
- The Top 10 routes confirm that Transjakarta is the backbone not just for DKI Jakarta, but for **Bodetabek** (Greater Jakarta) citizens.

Busiest Corridors



Insight:

- Maintain "**Direct Service**" routes during peak hours. Do not force passengers to transit at central hubs (like Cawang/Harmoni) as it increases travel time.
- Passengers from BSD, Bintaro, and Cibubur are upper-middle class car owners. Ensure **Park & Ride** facilities at starting points are expanded and safe. If parking is difficult, they will revert to using private cars.



DASHBOARD TRANSJAKARTA APRIL 2023

Day
Wednesday
Tuesday
Thursday
Saturday
Monday
Friday

Card Bank
bni
brizzi
dki
emoney
fiaz
online

Age Category
Boomer
Gen X
Gen Z
Millennial

Gender
Female
Male



DASHBOARD TRANSJAKARTA APRIL 2023

Revenue

Rp94,628,000

Transaction

35,970

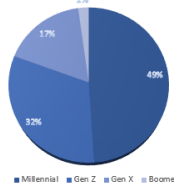
Avg. Ticket Price

Rp2,702.73

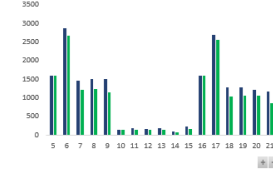
Missing Report

1,268

Passengers Profile

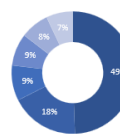


Travel Patterns by Male and Female



Financial

Revenue by Payment Method

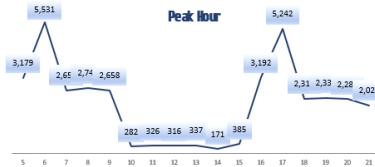


Pareto Revenue by Hour

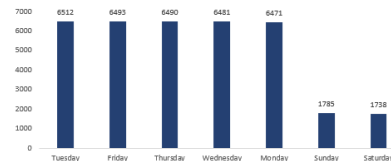


Peak Time & Busiest Corridors

Peak Hour



Peak Day



Busiest Corridors



Irfan Maulana

Data Analyst – Final Project Ms. Excel DQLab

Follow me:

Instagram: @i.ipaang

LinkedIn: <https://www.linkedin.com/in/maulana-irfan/>

Github: <https://github.com/fafnir96>

Thanks!